INCH-POUND

FF-S-86G
January 27, 2021
SUPERSEDING
FF-S-86F
January 7, 2020

# FEDERAL SPECIFICATION
# SCREW, CAP, SOCKET-HEAD

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers socket-head cap screws with multiple head configurations utilizing alloy steels (Grades 4037, 4137, 4140, 4142, 4145, 4340, 8740 and 5137M), corrosion-resistant steels (types 302, 302HQ, 304, 304L, 305, 316, 316L and 384), heat and corrosion-resistant steel (A286), and non-ferrous materials including nickel-copper alloy, nickel-copper-aluminum alloy, Inconel 625 or 686 and titanium alloys. Fasteners are furnished in plain, cadmium plating, zinc plating, phosphate coating, black chemical or passivated finishes. See the appropriate sections for limitations on use of each coating.

1.2 Classification.

1.2.1 Types and sizes. Screws furnished under this specification shall be of the following types (see 6.2 and Figure 7). Screws shall be classified according to size by the nominal body diameter as specified in ASME B18.3. Sizes covered range from #0 and up.

1/ DO NOT USE FOR NEW DESIGN (See 6.3.1)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to: General Services Administration, GSA Center (9FTE-10), Auburn, Washington 98001.

AMSC N/A FSC 5305

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NOTE: Some of the parts covered by this document are for replacement purposes only, and may not be included in new designs.

| | | |
|---|---|---|
| 1/ | Type I | - Hexagon socket Head Cap Screw series. |
| | Type IV | - Hexagon Socket Flat Countersunk Head Cap Screw with 82 Degree Included Angle only. |
| | Type VI | - Hexagon Socket Head Cap Screw |
| | Type VIII | - Hexagon Socket Button Head Cap Screw |

1.2.2 Part identification number system. A part identification number system shall be used to identify each fastener specified in this document. Part identification may use the numbering system called out on a non-government or military standard (e.g., NAS1351-02-8 or NASM1351C04H12, or MS24672-1), or the supplier may optionally use ASME B18.24, but they must specify which part number identification system is being used, as noted in 6.2.

1.3 Terminology. Definitions of terms used in this specification shall be as specified in ASME B18.12, unless otherwise defined herein.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL SPECIFICATIONS

| | |
|---|---|
| QQ-N-281 | – Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections. |
| QQ-N-286 | – Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500) |

FEDERAL STANDARDS

| | |
|---|---|
| FED-STD-H28/2 | – Screw-Thread Standards for Federal Services, Section 2, Unified Inch Screw threads- UN and UNR Thread Forms. |

MILITARY SPECIFICATIONS

| | |
|---|---|
| MIL-DTL-13924 | – Coating, Oxide, Black, for Ferrous Materials. |
| MIL-DTL-16232 | – Phosphate Coating, Heavy, Manganese or Zinc Base |
| MIL-F-495 | – Finish, Chemical, Black, for Copper Alloys. |



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MILITARY HANDBOOKS

MIL-HDBK-57 – Listing of Fastener Manufacturer's Identification Symbols

(Copies of the above documents are available online at https://quicksearch.dla.mil)

NAVSEA Technical Publications

T9074-AS-GIB-010/271 – Requirements for Nondestructive Testing Methods

(Copies of this document may be obtained from the Naval Sea Systems Command Metals, Welding and Fabrication Technical Publications Website – http://ntpdb.ddlomni.com/)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Quality (ASQ)

ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Applications for copies should be addressed to American Society for Quality, 60 North Plankinton Ave., Milwaukee, WI 53203 or http://asq.org)

ASTM International

| | |
|---|---|
| ASTM A380/A380M | – Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment and Systems, Standard Practice for |
| ASTM A493 | – Stainless Steel Wire and Wire Rods for Cold Heading And Cold Forging, Standard Specification for |
| ASTM A574 | – Alloy Steel Socket-Head Cap Screws, Standard Specification for |
| ASTM B21/B21M | – Naval Brass, Rod, Bar and Shapes, Standard Specification for |
| ASTM B98/B98M | – Copper-Silicon Alloy Rod, Bars and Shapes, Standard Specification for |
| ASTM B99/B99M | – Copper-Silicon Alloy Wire for General Applications, Standard Specification for |
| ASTM B138/B138M | – Manganese Bronze Rod, Bar and Shapes, Standard Specification for |



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| | |
|---|---|
| ASTM B139/B139M | – Phosphor Bronze Rod, Bar and Shapes, Standard Specification for |
| ASTM B150/B150M | – Aluminum Bronze Rod, Bar and Shapes, Standard Specification for |
| ASTM B348/B348M | – Titanium and Titanium Alloy Bars and Billets, Standard Specification for |
| ASTM B446 | – Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625), Nickel-Chromium-Molybdenum-Silicon Alloy (UNS N06219), and Nickel-Chromium-Molybdenum-Tungsten Alloy (UNS N06650) Rod and Bar, Standard Specification for |
| ASTM B574 | – Low-Carbon Nickel-Chromium-Molybdenum, Low-Carbon Nickel-Molybdenum-Chromium, Low-Carbon Nickel-Molybdenum-Chromium-Tantalum, Low-Carbon Nickel-Chromium-Molybdenum-Copper, and Low-Carbon Nickel-Chromium-Molybdenum-Tungsten Alloy Rod, Standard Specification for |
| ASTM E29 | – Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications |
| ASTM E112 | – Determining Average Grain Size, Standard Test Methods for |
| ASTM E384 | – Standard Test Method for Microindentation Hardness Of Materials |
| ASTM E1268 | – Assessing the Degree of Banding or Orientation of Microstructures, Standard Practice for |
| ASTM E1282 | – Specifying the Chemical Compositions and Selecting Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials, Standard Guide for |
| ASTM E1417/E1417M | – Liquid Penetrant Testing, Standard Practice For |
| ASTM E1444/E1444M | – Magnetic, Particle Testing, Standard Practice for |
| ASTM F468 | – Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use, Standard Specification for |
| ASTM F606/F606M | – Determining the Mechanical Properties of Externally and Internally Threaded Fasteners, Washers, Direct Tension Indicators, and Rivets, Standard Test Methods for |
| ASTM F788 | – Surface Discontinuities of Bolts, Screws, and Studs Inch and Metric Series, Standard Specification for |
| ASTM F835 | – Alloy Steel Socket Button and Flat Countersunk Head Cap Screws, Standard Specification for |
| ASTM F1941/F1941M | – Electrodeposited Coatings on Mechanical Fasteners Inch and Metric, Standard Specification for |

(Applications for copies should be addressed to ASTM, 100 Barr Harbor Drive, PO



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Box C700, West Conshohocken, PA 19428-2959 or https://www.astm.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

| | |
|---|---|
| ASME B1.1 | – Unified Inch Screw Threads (UN and UNR Thread Form) |
| ASME B1.3 | – Screw Thread Gaging Systems for Acceptability; Inch And Metric Screw Threads (UN, UNR, UNJ, M and MJ) |
| ASME B18.3 | – Socket Cap, Shoulder, Set Screws, and Hex Keys (Inch Series) |
| ASME B18.12 | – Terms for Mechanical Fasteners, Glossary of |
| ASME B18.18 | – Quality Assurance For Fasteners |
| ASME B18.24 | – Part Identifying Number (PIN) Code System Standard for B18 Fastener Product |
| ASME B46.1 | – Surface Texture (Surface Roughness, Waviness, and Lay) |

(Copies of these documents may be purchased from ASME, Two Park Avenue, New York, NY 10016-5990. http://www.asme.org/)

SAE International

| | |
|---|---|
| SAE-AMS2488 | – Anodic Treatment – Titanium and Titanium Alloys Solution pH 13 or Higher |
| SAE-AMS2700 | – Passivation of Corrosion Resistant Steels |
| SAE-AMS2759/1 | – Heat Treatment of Carbon and Low Alloy Steel Parts Minimum Tensile Strength Below 220 KSI (1517 MPA) |
| SAE-AMS5731 | – Steel, Corrosion and Heat Resistant, Bars, Wire, Forgings, Tubing and Rings 15CR - 25.5Ni - 1.2Mo – 2.1Ti - 0.006B - 0.30V Consumable Electrode Melted, 1800° F (982° C) Solution Heat Treated |
| SAE-AMS5853 | – Steel, Corrosion and Heat Resistant, Bars and Wire 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B 0.30V, Consumable Electrode Melted 1800°F (982°C) Solution Treated and Work-Strengthened 160ksi (1103 MPa) Tensile Strength |
| SAE-AMS-C-81562 | – Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited) |
| SAE-AMS-QQ-P-416 | – Plating, Cadmium (Electrodeposited) |
| SAE-J2295 | – Fasteners Part Standard – Cap Screws, Hex, Heavy Hex and Heavy Hex Structural Bolts, and Hex Nuts (Inch Dimensioned) |



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(Copies of these documents may be purchased from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001. http://www.sae.org/)

3. REQUIREMENTS:

3.1 Material.

3.1.1 Alloy Steel. Alloy steel screws shall be of a chemical composition conforming to Table I below, and when the specific alloy noted below is used, it shall be identified with the proper marking. Alloys with a minimum boron requirement are prohibited. The screws shall be fabricated from alloy steel made to fine grain practice as per ASTM E112. Unless otherwise noted in the procurement, fine grain size shall be defined as sizes below 100 nm or 0.1 micron. In the event of controversy over grain size, referee tests on finished screws shall be verified in accordance with test methods per ASTM E112.

Table I

| Grade Designation | 4037 | 4137 | 4140 | 4142 | 4145 | 4340 | 8740 | 5137M |
|---|---|---|---|---|---|---|---|---|
| UNS Number | G40370 | G41370 | G41400 | G41420 | G41450 | G43400 | G87400 | |
| Carbon: | | | | | | | | |
| Heat Analysis | 0.35-0.40 | 0.35-0.40 | 0.38-0.43 | 0.40-0.45 | 0.43-0.48 | 0.38-0.43 | 0.38-0.43 | 0.35-0.40 |
| Product Analysis | 0.33-0.42 | 0.33-042 | 0.36-0.45 | 0.38-0.47 | 0.41-0.50 | 0.36-0.45 | 0.36-0.45 | 0.33-0.42 |
| Manganese: | | | | | | | | |
| Heat Analysis | 0.70-0.90 | 0.70-0.90 | 0.75-1.00 | 0.75-1.00 | 0.75-1.00 | 0.60-0.80 | 0.75-1.00 | 0.30-0.50 |
| Product Analysis | 0.67-0.93 | 0.67-0.93 | 0.72-1.03 | 0.71-1.04 | 0.71-1.104 | 0.57-0.83 | 0.71-1.04 | 0.27-0.53 |
| Phosphorus, max.: | | | | | | | | |
| Heat Analysis | 0.035 | 0.035 | 0.035 | 0.0.35 | 0.035 | 0.035 | 0.035 | 0.035 |
| Product Analysis | 0.040 | 0.040 | 0.040 | 0.0.40 | 0.040 | 0.040 | 0.040 | 0.040 |
| Sulphur, max.: | | | | | | | | |
| Heat Analysis | 0.050 | 0.040 | 0.040 | 0.0.40 | 0.040 | 0.040 | 0.040 | 0.040 |
| Product Analysis | 0.055 | 0.045 | 0.045 | 0.045 | 0.045 | 0.045 | 0.045 | 0.045 |
| Silicon: | | | | | | | | |
| Heat Analysis | 0.20-0.35 | 0.15-0.35 | 0.15-0.30 | 0.15-0.35 | 0.15-0.35 | 0.15-0.35 | 0.15-0.35 | 0.15-0.35 |
| Product Analysis | 0.18-0.37 | 0.13-0.37 | 0.13-0.37 | 0.13-0.37 | 0.13-0.37 | 0.13-0.37 | 0.13-0.37 | 0.13-0.37 |
| Nickel: | | | | | | | | |
| Heat Analysis | *A* | *A* | *A* | *A* | *A* | 1.65-2.00 | 0.40-0.70 | *A* |
| Product Analysis | | | | | | 1.65-2.05 | 0.37-0.73 | |
| Chromium: | | | | | | | | |
| Heat Analysis | *A* | 0.80-1.10 | 0.80-1.10 | 0.80-1.10 | 0.80-1.10 | 0.70-0.90 | 0.40-0.60 | 0.90-1.20 |
| Product Analysis | | 0.75-1.15 | 0.75-1.15 | 0.75-1.15 | 0.75-1.15 | 0.67-0.93 | 0.37-0.63 | 0.85-1.25 |
| Molybdenum: | | | | | | | | |
| Heat Analysis | 0.20-0.30 | 0.15-0.25 | 0.15-0.25 | 0.15-0.25 | 0.15-0.25 | 0.20-0.30 | 0.20-0.30 | *A* |
| Product Analysis | 0.18-0.32 | 0.13-0.27 | | 0.13-0.27 | 0.13-0.27 | 0.18-0.32 | 0.18-0.32 | |
| Boron: | | | | | | | | |
| Heat Analysis | *A* | *A* | *A* | *A* | *A* | *A* | *A* | *A* |
| Product Analysis | | | | | | | | |

*A Elements shown with an "A" are not applicable to that grade designation.*

3.1.2 Corrosion-resistant steel (austenitic). Austenitic corrosion-resistant steel screws shall be manufactured from Type 302 (UNS S30200), Type 304 (UNS S30400), Type



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304L (UNS S30403), 302HQ (UNS30430), Type 305 (UNS S30500), Type 316 (UNS S31600), Type 316L (UNS S31603) or Type 384 (UNS S38400) , in accordance with the chemical compositions specified in ASTM A493 and the mechanical properties as specified in 3.2. Heat treatment and condition shall be in accordance with ASTM A493.

3.1.3 Heat and corrosion-resistant steel. Heat and corrosion-resistant steel screws shall be manufactured from A286 Iron Base Super alloy (UNS S66286) in accordance with AMS5731 or AMS5853. Cold work and precipitation hardening may be used to meet the mechanical properties specified in 3.2.

3.1.4 Non-ferrous material. Non-ferrous screws shall be manufactured from an alloy conforming to the applicable compositions specified in Table VI, and shall be heat treated per the appropriate document, so as to meet the requirements specified in Tables XV-XVII.

3.1.4.1 Grade 625 fasteners: The metal used in the production of the bar, rod, or wire used to make grade 625 fasteners shall be refined using the electroslag remelting process (ESR) or the vacuum arc remelting process (VAR). It must then be annealed after all manufacturing processes that can induce residual stresses are completed.

3.1.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used, in the production of the bar, rod, and wire used to make these fasteners, to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.1.6 Banding. Heavily banded microstructures, as described in ASTM E1268, are not permitted. This requirement applies only to alloy steels, and is not applicable to austenitic steels. See 4.3.6.

3.2 Mechanical properties. The ultimate tensile strength, proof stress, 0.2% offset yield strength, elongation, reduction of area, and hardness requirements of finished fasteners shall be IAW Tables XV and XVII. Machined specimens shall follow the requirements of Table XVII.

3.2.1 Corrosion-resistant steel screws. Corrosion-resistant steel screws shall be furnished in the annealed condition unless the cold worked condition is specifically called out in the ordering date.

3.2.1.1 Cold worked condition. The mechanical properties listed herein for cold worked corrosion resistant steel screws are valid for sizes up to and including 0.625 inches in diameter. Mechanical properties for cold worked corrosion resistant steel screws shall be agree upon by the user and the manufacturer for sizes over 0.625 inches in diameter.



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3.2.2 All screws. Screws with hardness values below the minimum requirement may be accepted provided they pass all of the other requirements. Fasteners that exceed the maximum required hardness shall not be offered for acceptance.

3.3 Finish. Unless otherwise specified in the part standard, screws shall be furnished uncoated or with a protective coating or surface treatment as specified herein (see 6.2 and 6.3). If no coating is specified, alloy steel screws shall have a black oxide coating.

3.3.1 Passivation.

3.3.1.1 Corrosion-resistant steel cap screws shall be cleaned, descaled and passivated in accordance with ASTM A380/380M, AMS2700, Method 1, Type 2, Type 6 or Type 7, or AMS2700, Method 2.

3.3.1.2 Heat and corrosion-resistant steel cap screws shall be passivated in accordance with AMS2700, Method 1, Type 2 or Type 8, Class 1.

3.3.2 Cadmium Plating (Obsolete for New Design, see 6.3.2). When specified, screws shall be cadmium plated by electrodepositing in accordance with AMS-QQ-P-416, Type II, Class 3 or by mechanical plating in accordance with AMS-C-81562, Type II, Class 3. For Navy and Air Force acquisition only, when specified, AMS-QQ-P-416, Type II, Class 2 or AMS-C-81562, Type II, Class 2 shall apply. When cadmium plating is specified for screws of 48 pitch or finer, plating thickness may be reduced below .0002 inch (5µm). Supplementary chromate treatment shall be accomplished by a method that does not allow hexavalent chromium to be produced. (See 6.3.3.)

3.3.3 Zinc coating. When specified, screws shall be zinc coated by electrodepositing in accordance with ASTM F1941/F1941M, Class Fe/Zn 5 or by mechanical plating in accordance with AMS-C-81562, Type II, Class 6, except all alloy steel fasteners shall receive hydrogen embrittlement relief, regardless of hardness. Supplementary chromate treatment shall be accomplished by a method that does not allow hexavalent chromium to be produced. Do not allow supplementary treatments on a zinc coating that is less than 5 µm thick. For zinc coating in accordance with ASTM F1941/F1941M, the fasteners shall receive a 23 hour bake at 375 degrees F +/- 25 degrees F initiated within one hour after removal from the plating bath for hydrogen embrittlement relief treatment. When zinc coating is specified for screws of 48 pitch or finer, plating thickness may be reduced below .0002 inch (5 µm). A 96 hour salt spray test shall be performed in accordance with ASTM F1941/F1941M for Class Fe/Zn-5. Alloy steel fasteners that are coated shall verify elongation. (See 3.2.1.1 (d) and Note 2 to Table VI).

3.3.4 Phosphate coating. When specified, screws shall be phosphate coated in accordance with MIL-DTL-16232, Type Z, Class 2.



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3.3.5 Black chemical finish. When specified, black oxide coatings shall be applied as specified in 3.3.5.1, 3.2.5.2, 3.3.5.3 and 3.3.5.4 below.

3.3.5.1 Alloy steel screws. When specified, alloy steel screws shall have a black chemical finish in accordance with MIL-DTL-13924 Class 4, or MIL-DTL-16232, Type M, Class 4. Fasteners .2500 and larger will be marked with their alloy.

3.3.5.2 Corrosion-resistant steel screws. When specified, corrosion-resistant steel screws shall have a black chemical finish in accordance with MIL-DTL-13924 Class 4 or an equivalent method that does not utilize hexavalent chromium and meets the same performance requirements. Fasteners .2500 and larger will be marked with their alloy.

3.3.5.3 Heat and corrosion-resistant steel screws. When specified, corrosion-resistant steel screws shall have a black chemical finish in accordance with MIL-DTL-13924 Class 3 or an equivalent method that does not utilize hexavalent chromium and meets the same performance requirements. Fasteners .2500 and larger will be marked with their alloy.

3.3.5.4 Copper alloy screws. When specified, copper alloy screws shall have a black chemical finish in accordance with MIL-F-495 or an equivalent method that does not utilize hexavalent chromium and meets the same performance requirements. Fasteners .2500 and larger will be marked with their alloy.

3.3.6 Non-ferrous screws (other than copper and titanium alloy). Non-ferrous screws, other than copper and titanium alloy, shall not receive a black chemical finish.

3.3.7 Titanium Screws. When specified, titanium screws shall have anodic treatment in accordance with AMS2488, Type II.

3.4 Surface texture. For alloy steel cap screws of sizes up to and including .625 inch, and having nominal lengths equal to or less than 8 times the basic screw diameter, the surface roughness of the screws before plating shall not exceed 63 microinches RA on the fillet and head bearing surfaces, nor exceed 32 microinches on the threads. For larger sizes, longer lengths, and corrosion resistant steel screws, the surface roughness of the screws before plating shall not exceed 125 microinches RA on the body, fillet and head bearing surfaces. Normally, it will be sufficient to ascertain that these surfaces on screws have the equivalent of a smooth machined finish by visual comparison with known surface standards. However, when it is deemed necessary to measure these surfaces with commercially available equipment, roughness measurements shall be taken axially on the body and fillet surfaces, and circumferentially on the bearing surface in accordance with ASME B46.1.

3.5 Design.

3.5.1 Dimensions. Unless otherwise specified, screws shall conform to the dimensions



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and tolerances specified in ASME B18.3. Exceptions must be noted per Section 6.2 (Procurement options).

3.5.2 Cylindrical heads (drilled). When specified (see 6.2 and 6.3) screws shall be cross drilled in accordance with Table II (below). The alignment plug specified in Table II shall pass completely through the head without deflection. On nominal sizes .164 inch and above, the drilled holes shall not break through the corners of the hexagon socket. Nominal sizes .112 inch and .138 inch shall have two drilled holes spaced 180°; sizes.164 inch and above shall have six drilled holes spaced 60°. Edges of holes in the sockets may contain burrs; however, the socket shall accept a key in accordance with ASME B18.3.

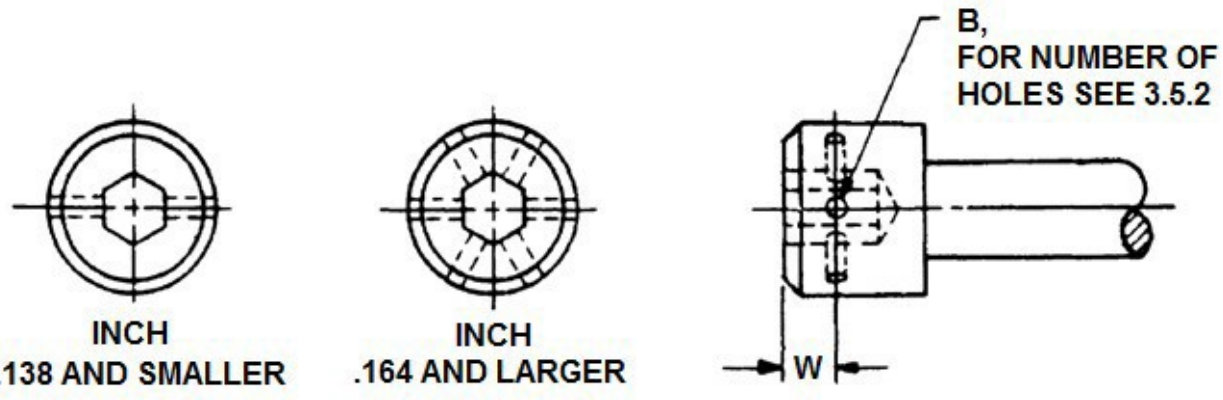
Table II. Dimensions for drilled-head socket screws.

| NOMINAL SIZE | W TOP OF HEAD TO CENTER OF HOLE | | B DRILLED HOLE DIAMETER | | HOLE ALIGNMENT PLUG DIAMETER |
|---|---|---|---|---|---|
| | MAX | MIN | MAX | MIN | |
| .1120 | .040 | .026 | .039 | .033 | .025 |
| .1380 | .050 | .035 | .039 | .033 | .025 |
| .1640 | .060 | .040 | .050 | .044 | .030 |
| .1900 | .065 | .045 | .050 | .044 | .030 |
| .2500 | .085 | .065 | .050 | .044 | .030 |
| .3125 | .104 | .084 | .050 | .044 | .030 |
| .3750 | .123 | .103 | .067 | .061 | .051 |
| .4375 | .141 | .121 | .067 | .061 | .051 |
| .5000 | .160 | .140 | .067 | .061 | .051 |
| .6250 | .198 | .178 | .067 | .061 | .051 |
| .7500 | .235 | .215 | .097 | .091 | .081 |
| .8750 | .273 | .253 | .097 | .091 | .081 |
| 1.0000 | .310 | .290 | .097 | .091 | .081 |
| 1.1250 | .348 | .328 | .127 | .119 | .104 |
| 1.2500 | .385 | .365 | .127 | .119 | .104 |
| 1.3750 | .423 | .403 | .127 | .119 | .104 |
| 1.5000 | .460 | .440 | .127 | .119 | .104 |

3.5.3 Broaching chips. Broached sockets of screws nominal size .112 or larger shall have all loose chips removed. Remaining chips shall be firmly attached.



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3.5.4 Threads.

3.5.4.1 Method of manufacture of threads on fasteners (except Ni-Cu-Al). Screw threads on fasteners shall be rolled for diameters through .625 inch and for screw lengths through 4.000 inch. For diameters and lengths other than this, threads shall be rolled, cut or ground.

3.5.4.2 Method of manufacture of threads on Nickel-Copper-Aluminum fasteners (Ni-Cu-Al, K-Monel). Ni-Cu-Al material that has been headed or roll threaded shall not be age hardened unless it has been solution annealed subsequent to the heading and threading operations. Threads formed after the final age-hardening heat treatment shall be cut or ground.

3.5.4.3 Thread series and class. Unless otherwise specified (see 6.2 and 6.3) threads shall be UNC, UNRC, UNF, UNRF or UN8 series: Typical Class 3A for nominal diameter sizes .060 through 1.000 inch inclusive and Class 2A for nominal diameter sizes over 1.000 inch and larger in accordance with FED-STD-H28/2. When applying coating to Class 2A, the maximum material limit of Class 3A shall be met after application. Type VI shall be UNRC or UNRF series.

3.5.5 Source identification mark. Screws with nominal size .1900 and larger shall be permanently marked to identify the source accepting responsibility for the screws meeting all requirements specified herein, including coatings. The marking shall be a source identifying symbol for a manufacturer in accordance with in MIL-HDBK-57 or a private label distributor's symbol as applicable, and must not be on a bearing surface.

3.5.6 Material Identification Mark. Screws with nominal size .2500 and larger shall be permanently marked to identify the alloy/mechanical property specified in 3.2. The marking shall be raised or depressed at the option of the manufacturer and located on the top or the cylindrical side of the screw head such that the marking is visible after installation and any applied coating, unless otherwise specified on the applicable specification sheet. If there is not enough room to mark both the alloy and condition,



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then the alloy marking shall take precedence. All material and condition markings are designated in Table VI.

3.6 Carburization and decarburization limits for alloy steel screws. For alloy steel screws, the extent of carburization and decarburization shall be in accordance with the requirements of ASTM A574 and ASTM F835, as applicable.

3.7 Discontinuities. Surface discontinuities for these products shall conform to ASTM F788 with supplemental requirement S1 and the additional limitations specified below.

3.7.1 Threads. Threads shall have no laps at the root or on the flanks below the pitch diameter line as shown in Figure 1. Laps are permissible at the crest to a depth of 25 percent of the basic thread height and on the flanks above the pitch diameter line as shown in Figure 2. Slight deviation from the thread contour is permissible at the crest of the thread as shown in Figure 3. Longitudinal seams rolled beneath the root of the thread and longitudinal seams in the crest of cut threads are acceptable within the limits of Table III, column 1 (below).

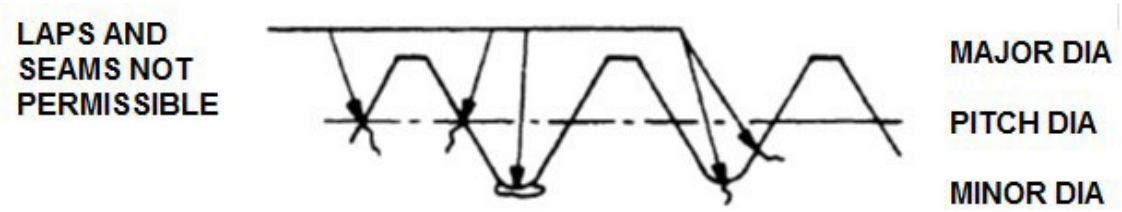



Figure 1. Discontinuities below pitch diameter line.

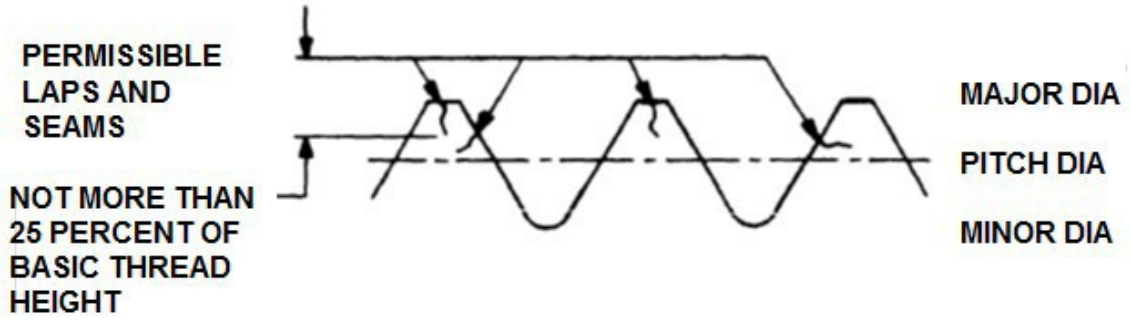



Figure 2. Discontinuities above pitch diameter.



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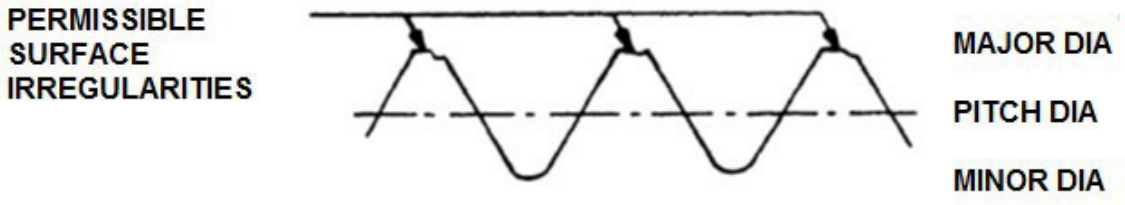



Figure 3. Discontinuities in crest contour.

Table III. Discontinuity Limits

| Nominal Size | Permissible Discontinuity Depth (max) Bearing, Fillet and Body Surfaces 1/ | Permissible Discontinuity Depth (max) Head Surfaces 2/ |
|---|---|---|
| .0600 | .005 | .007 |
| .0730 | .005 | .007 |
| .0860 | .005 | .007 |
| .0990 | .005 | .007 |
| .1120 | .005 | .007 |
| .1380 | .005 | .008 |
| .1640 | .005 | .010 |
| .1900 | .006 | .011 |
| .2500 | .008 | .015 |
| .3125 | .009 | .018 |
| .3750 | .011 | .023 |
| .4375 | .013 | .026 |
| .5000 | .015 | .030 |
| .5625 | .017 | .034 |
| .6250 | .019 | .038 |
| .7500 | .023 | .045 |
| .8750 | .026 | .053 |
| 1.0000 | .030 | .060 |
| 1.1250 | .034 | .064 |
| 1.2500 | .038 | .064 |
| 1.3750 | .041 | .064 |
| 1.5000 | .045 | .064 |

NOTE: Formulas for permissible discontinuity depths:
1/ Formula 1: Bearing, area, fillet, and other surfaces- max depth = .03D or .005 inch (whichever is greater).



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2/ Formula 2: Peripheral discontinuities - max depth = .06D, but not to exceed .064 inch.

NOTE: D = nominal diameter of screw.

3.7.2 Cylindrical head screws. The limits of acceptable discontinuities for cylindrical head screws shall be in accordance with Figure 4, and Table III, on all surfaces except threads and fillets. Discontinuities as defined in 6.4, are permitted in the locations illustrated in Figure 4 to the depths shown. All discontinuities are to be measured perpendicular to the indicated surfaces. The socket discontinuity locations and limits shall be as specified in ASTM A574.

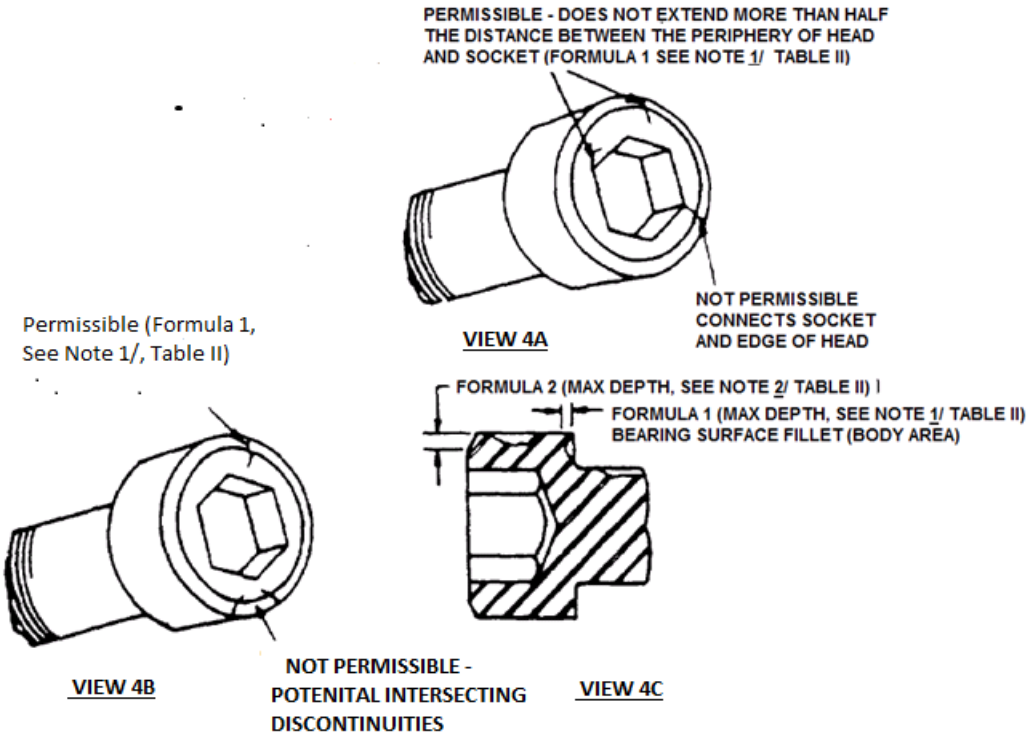



Figure 4. Discontinuities in cylindrical heads.

3.7.3 Flat countersunk head screws (82 degree included angle). The limits of acceptable discontinuities for flat countersunk head screws shall be in accordance with Figure 5 on all surfaces except threads and fillets. The socket discontinuity locations and limits shall be as specified in ASTM A574.



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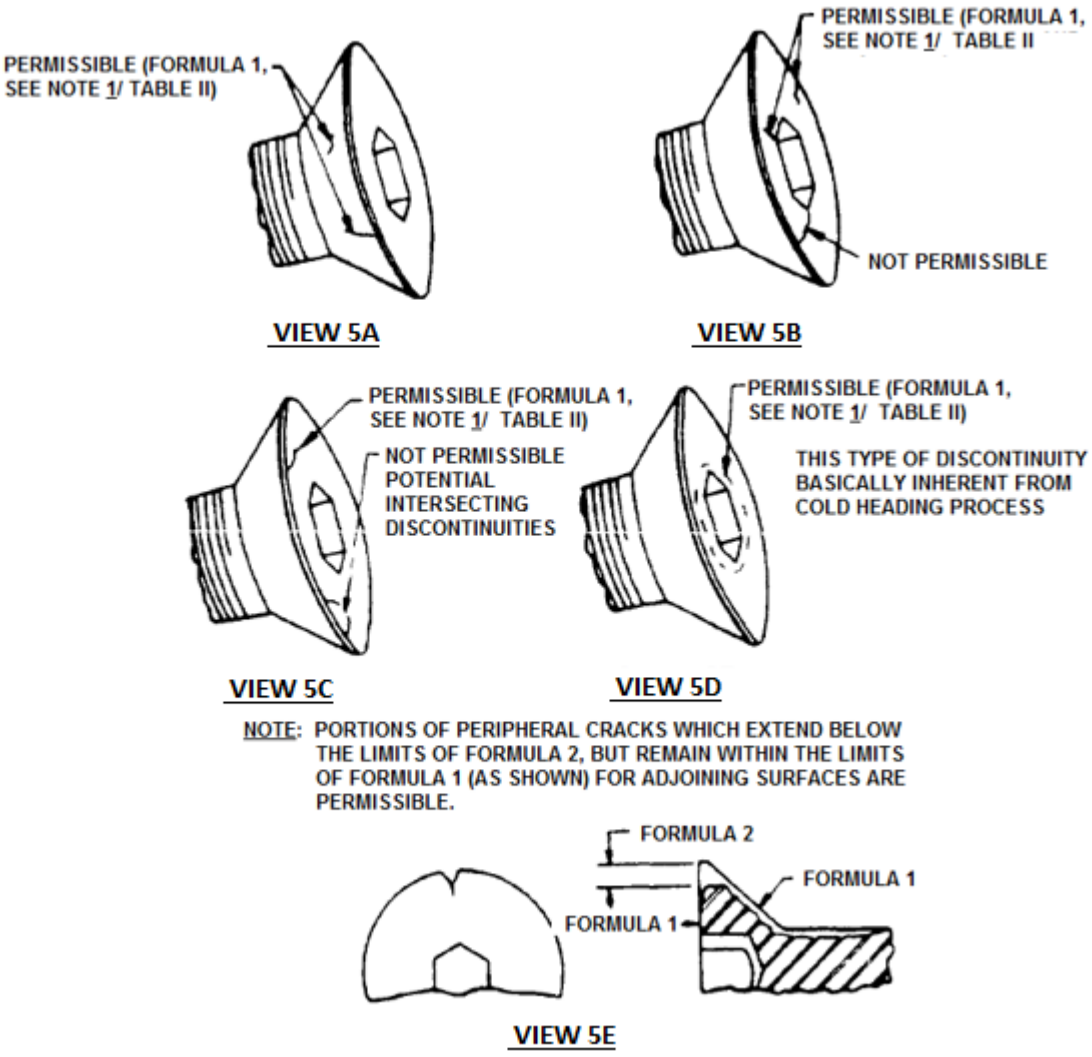



Figure 5. Discontinuities in flat countersunk heads.

3.7.4 Button head screws. The limits of acceptable discontinuities for button head screws shall be in accordance with Figure 6 on all surfaces except threads and fillets. The socket discontinuity locations and limits shall be as specified in ASTM A574.



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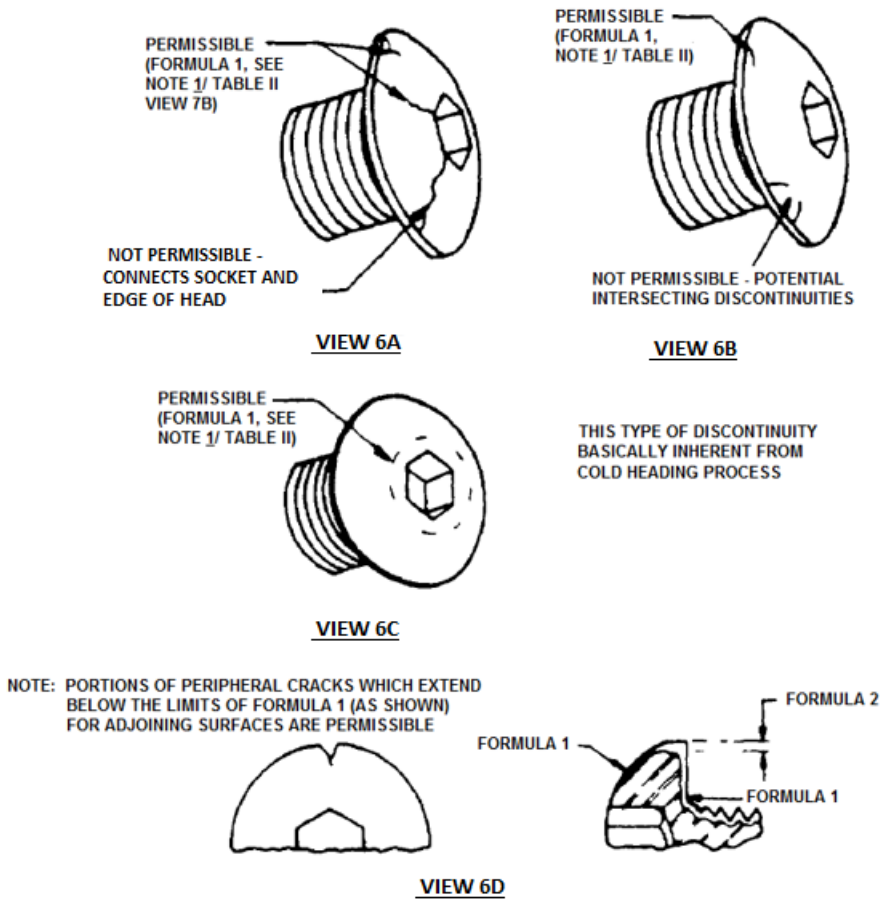



Figure 6. Discontinuities in button heads.

3.8 Workmanship. Screws shall be free from burrs, cracks, seams, laps, nicks, pits, loose scale, irregular surfaces, chips and other defects that will adversely affect life or serviceability.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the manufacturer is responsible for the performance of all inspection requirements as specified herein. (See 6.2.) Except as otherwise specified in the contract, the manufacturer may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.



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4.2.1 Lot. A lot shall consist of all screws of the same type, made from the same heat of material, heat treatment, protective coating, surface finish, size, length, and threads produced and manufactured at the same time by the same manufacturer's facility and offered for acceptance at the same time. Where the length of the lot of fasteners is too short to support mechanical testing of full size fasteners per 4.3, the lot shall include longer length fasteners solely for the purposes of mechanical testing. (See 4.2.3 and 4.2.4 for direction regarding testing of too short or too long fasteners). These fasteners shall be identical (material, type, etc.) to those fasteners in the lot it represents, except for their overall length and number of threads. See 4.3.1 for the maximum length of the longer fasteners. The longer test fasteners shall be made from the same heat of rod, bar, or wire used for the production fasteners; have the same type, style, and diameter as the production fasteners; and shall be processed and heat treated with the production fasteners. For non-plated part numbers cut off from a previously tested longer part number, the previous part number's mechanical testing can be used in lieu of test bolts. Current inventory manufactured to FF-S-86 Revision G which is too short for testing may be certified to revision G with the agreement of the contracting officer and test results proving compliance with all requirements. K-Monel FF-S-86E fasteners must provide certification that threads are cut vice rolled.

4.2.2 Sampling for examination. A random sample of screws shall be taken from each lot in accordance with ASQ Z1.4, Table IV-A, using multiple sampling plans, unless otherwise specified by the contract. For small lot procurement, see 4.2.9.

4.2.3 Sampling for fasteners too short for testing. For lots with fasteners that are too short for testing in accordance with ASTM F606/F606M, the longer length fasteners manufactured with the same lot to support mechanical testing shall be utilized to support proof, yield, axial and wedge strength testing, but shall not be used for evaluation of overall screw length, thread length or grip length.

4.2.4 Sampling for fasteners too long for testing. For lots of fasteners too long for testing in accordance with ASTM F606/F606M, standard length fasteners manufactured with the same lot to support mechanical testing shall be utilized to support proof, yield, axial and wedge strength testing, but shall not be used for evaluation of overall screw length, thread length or grip length.

4.2.5 Sampling for mechanical properties and metallographic tests. A random sample of screws shall be taken from each lot in accordance with ASQ Z1.4, Table IV-A, using multiple sampling plans, unless otherwise specified by the contract for destructive testing. (See 6.2.) . For small lot procurement, see 4.2.7. For lots of fasteners that are too short for testing in accordance with ASTM F606/F606M, see 4.2.3 and 4.2.4, respectively. Fasteners that have been yield strength tested in accordance with ASTM F606/F606M utilizing either Method 2 or Method 2A shall not be used for wedge testing.

4.2.6 Sampling for protective coating or surface treatment tests. Unless otherwise specified in the relevant procedure specification, sampling for tests of protective coating or surface treatment shall be in accordance with the following Table VI:



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Table IV. Lot Sample Requirements

| Lot Size | Number of Samples | Accept | Reject |
|---|---|---|---|
| Less than 151 | 1 | 0 | 1 |
| 151 to 10,000 | 2 | 0 | 1 |
| 10,0001 to 35,000 | 3 | 0 | 1 |
| Greater than 35,000 | 4 | 0 | 1 |

Table V. Classification of defects

| CATEGORIES | DEFECTS | INSPECTION METHOD |
|---|---|---|
| CRITICAL<br>1<br>2<br>3 | AQL = 1.00<br>Discontinuities (see 3.7)<br>Source identification mark (see 3.5.5)<br>Material Marking (except alloy steel) (See 3.5.6) | <br>CIE 1/<br>Visual<br>Visual |
| Major<br>101<br>102 | AQL = 2.50 Socket<br>dimensions (see 3.5.1)<br>Threads (see 3.5.4) | <br>CIE 1/<br>CIE 1/ |
| Minor | AQL = 4.00 | |
| 201 | Protective coating or surface treatment (see 3. 3) | Visual/CIE 1/ |
| 202 | Surface roughness (see 3.4) | Visual/CIE 1/ |
| 203 | Body diameter (cylindrical and flat countersunk heads) (see 3.5.1) | CIE 1/ |
| 204 | Head diameter (see 3.5.1) | CIE 1/ |
| 205 | Head height (see 3.5.1) | CIE 1/ |
| 206 | Head chamfer (see 3.5.1) | CIE 1/ |
| 207 | Bearing surface (see 3.5.1) | CIE 1/ |
| 208 | Edge of head radius or chamfer (cylindrical heads) (see 3.5.1) | CIE 1/ |
| 209 | Concentricity (see 3.5.1) | CIE 1/ |
| 210 | Under head fillet (see 3.5.1) | CIE 1/ |
| 211 | Screw length (see 3.5.1) | CIE 1/ |
| 212 | Thread length (see 3 5 .1) | CIE 1/ |
| 213 | Grip gaging length (cylindrical and flat countersunk heads) (see 3.5.1) | CIE 1/ |
| 214 | Screw point chamfer (see 3.5.1) | CIE 1/ |
| 215 | Flushness tolerance (flat countersunk heads) (see 3.5.1) | CIE 1/ |
| 216 | Drilled hole dimensions (cylindrical heads) (see 3.5.2) | CIE 1/ |
| 217 | Broaching chips (see 3.5.3) | Visual/CIE 1/ |

NOTE: 1/ Commercial Inspection Equipment



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4.2.7 Sampling for chemical analysis. A random sample of screws shall be taken from each lot in accordance with ASQ Z1.4, Table IV-A ,using multiple sampling plans, unless otherwise specified by the contact for destructive testing. (See 6.2.) The acceptable quality level (AQL) shall be 1.0. Samples previously selected for examination or for mechanical properties or metallographic tests may be used. See 4.3.4 for use of mill certs for chemical analysis. For small lot procurement, see 4.2.9.

4.2.8 Sampling for discontinuities. A random sample of screws shall be taken from each lot in accordance with Tables I and IV of ASQ Z1.4, Table IV-A, unless otherwise specified by the contact non-destructive testing. (See6.2.) The AQL shall be 1.0. For small lot procurement see 4.2.9.

4.2.9 Sampling for small lots. For small lot procurement of less than 51 screws, sampling and acceptance/rejection criteria for mechanical properties, metallographic, chemical analysis, and discontinuity tests shall be as noted in Table VI, below.

Table VI

| | | Number of Defectives | |
|---|---|---|---|
| Lot size[1] | Sample size | Accept | Reject |
| 2 to 50 | 3 | 0 | 1 |

1. For lot sizes of two (2) pieces, sample size shall be two (2).

4.2.10 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be as specified in Section 5.

4.2.11 Examination.

4.2.11.1 Visual and dimensional examination. Each screw taken as specified in 4.2.2 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table V and ASME B18.3. Any screw in the sample containing one or more defects shall be rejected, and if the number of defective screws in any sample exceeds the acceptance number for that category, the lot represented by the sample shall be rejected. Alternately, at the procuring agency's discretion, 100% inspection in accordance with ASME B18.18 may be permitted.

4.2.11.2 Thread inspection. Screw threads shall be inspected in accordance with ASME B1.3, System 22.

4.3 Test methods.

4.3.1 Mechanical property test. Mechanical tests shall be performed on each lot of fasteners per Table VIII based on material and head type. Screws shall be tension tested full size, when practical. To support full size testing of short length fasteners, longer fasteners produced from the same lot per 4.2.1 shall be utilized in lieu of tension testing a



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machined specimen. Substitution of longer length fasteners in the lot for short fasteners during tension testing shall be allowed only when the fastener length does not meet the minimum length requirement of ASTM F606/F606M. The maximum length of the longer fastener for testing shall be the limited to the minimum length cited in ASTM F606/F606M plus ¼ inch or 3D, whichever is longer. Gage length shall not be increased by machining threads into the shank.

TABLE VII. Chemical Requirements and Material Markings

| Grade | Material | Condition | Chemical Requirement | Alloy Marking | Condition Marking |
|---|---|---|---|---|---|
| AS | Alloy Steel | Not Specified | See 3.1.1 | NONE | NONE |
| SS300 | Corrosion Resistant Steel | Annealed (A) or Cold Worked (CW) | See 3.1.2 | "302" for AISI 302 | "A" or "CW" |
| | | | | "B8" for AISI 304 | A or "CW" |
| | | | | "304L" for AISI 304L | A or "CW" |
| | | | | "B8P" for AISI 305 | "A" or "CW" |
| | | | | "384" for AISI 384 | A or "CW" |
| SS316 | Corrosion Resistant Steel | Annealed (A) or Cold Worked (CW) | | "316" or "B8M" for AISI 316 | "A" or "CW" |
| | | | | "316L" for AISI 316L | "A" or "CW" |
| A286 | Heat and Corrosion Resistant Steel | Not Specified | See 3.1.3 | "A286" or "N" | NONE |
| 6575A | Manganese Bronze | O60 - Soft Anneal | ASTM F468 UNS C67500 | "675A" | NONE |
| 632A | Nickle Aluminum Bronze | TQ50 - Quench Hardened and Tempered | ASTM B150/B150M UNS C63200 | "632A" | NONE |
| 510A | Phosphor Bronze | H04 - Hard | ASTM F468 ASTMB139/B139M UNS C51000 | "510A" | NONE |
| 651A | Silicone Bronze | 04H - Hard or 01H - Quarter Hard | ASTM F468; ASTM B98/B98M; ASTM B99/B99M; UNS C65100 | "651A" | NONE |
| 464A | Naval Brass | H02 - Half Hard or O50 - Light Annealed | ASTM F468; ASTM B21/B21M; UNS C46400 | "464A" | NONE |
| N400 | Monel | Hot Finished or Cold Drawn | ASTM F468; QQ-N-281; UNS N04400/N04405 | "NICU" | NONE |
| N500 | K-Monel 500 | Annealed and Age Hardened | QQ-N-286 UNS N05500 | "•K•" | NONE |
| 625A | Inconel 625 | Annealed | ASTM F468; ASTM B446 Grade 1; UNS N06625 | "625A" | NONE |
| 686A | Inconel 686 | Hot or Cold Worked and Annealed | ASTM F468; ASTM B574; UNS N06686 | "686A" | NONE |
| T23 | Titanium | Hot Worked or Cold Finished | ASTM F468; ASTM B348; UNS R56407 | "T23" | NONE |



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| Table VIII. Mechanical Testing Requirements | | | | | | | |
|---|---|---|---|---|---|---|---|
| Material | Head Type (See 1.2.1) | Hardness (See 4.3.1.1) | Proof Test (See 4.3.1.2.1) | Yield Test (See 4.3.1.2.2) | Axial Tension Test (See 4.3.1.2.3) | Wedge Tension Test (See 4.3.1.3) | Elongation and Reduction In Area (See 4.3.1.2.4) |
| Alloy Steel | IV,VIII | X | X | | X | | (Note 3) |
| | VI | X | X | | | X | (Note 3) |
| | Machined Specimen (Note 1) | X | | X | X | | |
| CRES, & Heat and Corrosion Resistant Steel | IV,VIII | X | X | | X | | |
| | VI | X | | X | | X | |
| | Machined Specimen (Note 1) | X | | X | X | | X |
| Non-Ferrous | IV,VIII | X | X | | X | | |
| | VI | X | | X | | X | |
| | Machined Specimen (Note 1) | X | | X | X | | X |

Note 1: For testing screws having tensile properties greater than canbe measured on available test equipment, the body shall be turned down, concentric to the axis of the screw, to the largest practical gage dimension specified in ASTM F606/F606M. These specimens shall be tested for tensile strength, yield strength, elongation and reduction in area in accordance with ASTM F606/F606M. Non-ferrous fasteners shall be excluded from the reduction in area requirement.
Note 2: 0.2% offset shall be used for yield strength testing unless otherwise cited in a reference document.
Note 3: Elongation in 4D testing is required when alloy steel fasteners equal to or larger than ½" nominal diameter are to be electroplated (See 3.2.1.1(d)).
Note 4: Due to the critical nature of the fastener head for Types IV and VII socket head cap screws, the yield and tensile strength are reduced by 20%,
Note 5: Proof loads shall be 90% of the material's tensile strength.
Note 6: For arbitration purposes, ultimate tensile strength takes precedence over hardness.
Note 7: Tensile area shall be calculated IAW either FED-STD-H28 or ASME B1.1 (See Table VI).



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| Table IX Stress Areas for Standard Size Inch Fasteners | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Nominal Diameter (in) | | Coarse Threads | | | Fine Threads | | | Series 8UN Threads | | |
| | | Threads/ Inch | Tensile Area (in) | Shear Area (in) | Threads/ Inch | Tensile Area (in) | Shear Area (in) | Threads/ Inch | Tensile Area (in) | Shear Area (in) |
| 0.073 | #1 | 64 | 0.00263 | 0.00218 | 72 | 0.00278 | 0.00237 | | | |
| 0.086 | #2 | 56 | 0.00370 | 0.00310 | 64 | 0.00394 | 0.00339 | | | |
| 0.990 | #3 | 48 | 0.00487 | 0.00406 | 56 | 0.00523 | 0.00451 | | | |
| 0.112 | #4 | 40 | 0.00604 | 0.00496 | 48 | 0.00661 | 0.00566 | | | |
| 0.138 | #6 | 32 | 0.00909 | 0.00745 | 40 | 0.01015 | 0.00874 | | | |
| 0.164 | #8 | 32 | 0.0140 | 0.01196 | 36 | 0.01475 | 0.01285 | | | |
| 0.190 | #10 | 24 | 0.0175 | 0.0145 | 32 | 0.0200 | 0.0175 | | | |
| 0.250 | 1/4 | 20 | 0.0318 | 0.0269 | 28 | 0.0364 | 0.0326 | | | |
| 0.3125 | 5/16 | 18 | 0.0524 | 0.0454 | 24 | 0.0581 | 0.0524 | | | |
| 0.375 | 3/8 | 16 | 0.0775 | 0.0678 | 24 | 0.0878 | 0.0809 | | | |
| 0.4375 | 7/16 | 14 | 0.1063 | 0.0933 | 20 | 0.1187 | 0.1090 | | | |
| 0.500 | 1/2 | 13 | 0.1419 | 0.1257 | 20 | 0.1600 | 0.1486 | | | |
| 0.5625 | 9/16 | 12 | 0.1819 | 0.1621 | 18 | 0.2030 | 0.1888 | | | |
| 0.625 | 5/8 | 11 | 0.2260 | 0.2018 | 18 | 0.2560 | 0.2400 | | | |
| 0.750 | 3/4 | 10 | 0.3345 | 0.3020 | 16 | 0.3730 | 0.3513 | | | |
| 0.875 | 7/8 | 9 | 0.4617 | 0.4193 | 14 | 0.5095 | 0.4805 | | | |
| 1.000 | 1 | 8 | 0.6057 | 0.5510 | 12 | 0.6630 | 0.6246 | | | |
| 1.125 | 1 1/8 | 7 | 0.7633 | 0.6931 | 12 | 0.8557 | 0.8119 | 8 | 0.7905 | 0.7278 |
| 1.250 | 1 1/4 | 7 | 0.9691 | 0.8899 | 12 | 1.0729 | 1.0238 | 8 | 0.9997 | 0.9291 |
| 1.375 | 1 3/8 | 6 | 1.1549 | 1.0541 | 12 | 1.3147 | 1.2603 | 8 | 1.2335 | 1.1549 |
| 1.500 | 1 1/2 | 6 | 1.4053 | 1.2938 | 12 | 1.5810 | 1.5213 | 8 | 1.4918 | 1.4053 |
| 1.750 | 1 3/4 | 5 | 1.8995 | 1.7441 | | | | 8 | 2.0822 | 1.9796 |
| 2.000 | 2 | 4.5 | 2.4982 | 2.3001 | | | | 8 | 2.7706 | 2.6522 |
| 2.250 | 2 1/4 | 4.5 | 3.2477 | 3.0213 | | | | 8 | 3.5573 | 3.4229 |
| 2.500 | 2 1/2 | 4 | 3.9988 | 3.7162 | | | | 8 | 4.4421 | 4.2918 |

Note 1: Tensile area, $A_t$ and shear area, $A_s$ are determined by the following equations

$$A_t = 0.7854\left[D - \frac{0.9743}{n}\right]^2$$

$$A_s = \frac{\pi}{4}\left[D + \frac{1.2990381}{n}\right]^2$$

Where:
$A_t$ = Tensile stress area, $in^2$
$A_s$ = Shear stress area, $in^2$
D = Nominal size (basic major diameter), in
n = Number of threads per inch



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4.3.1.1 Hardness test. Samples selected as specified in 4.2.5 shall be subjected to a hardness test conducted in accordance with ASTM F606/F606M at room temperature on a clean, flat surface, capable of sustaining the load imposed by the hardness tester.

4.3.1.2 Axial test. Samples selected as specified in 4.2.5 shall be subjected to a full size tension test conducted in accordance with ASTM F606/F606M. For flat countersunk and button head screws, tensile failures through the head are acceptable provided the load requirements are satisfied.

4.3.1.2.1 Proof test. Samples selected as specified in 4.2.5 shall be subjected to proof testing of full-size fasteners in accordance with the length measurement method of ASTM F606/F606M. The alternative yield strength method in ASTM F606/F606M may be used for determining the ability of the fastener to withstand the proof test, but the proof test shall be the arbitration method. Proof load tested flat countersunk head and button head fasteners with deformed heads shall be considered to have failed the proof load test.

4.3.1.2.2 Yield test. Samples selected as specified in 4.2.5 shall be subjected to yield strength testing in accordance with ASTM F606/F606M. Fasteners that have been yield strength tested in accordance with ASTM F606/606M utilizing either Method 2 or Method 2A shall not be used for wedge testing.

4.3.1.2.3 Axial tension test. Samples selected as specified in 4.2.5 shall be subjected to axial tension testing in accordance with ASTM F606/F606M. For flat countersunk head and button head fasteners, types IV and VIII, the fracture may extend or spread into the head of the fastener. Minimum loads required to meet the tensile requirements are listed in Tables IX and XIII for corrosion resistant steel and heat and corrosion resistant steel screw only.

4.3.1.2.4 Tension testing of machined test specimen. For testing screws having tensile properties greater than can be measured on available test equipment, the body shall be turned down, concentric to the axis of the screw, to the largest practical gage dimension specified in ASTM F 606/F606M. These specimens shall be tested for tensile strength, yield strength, elongation and reduction in area in accordance with ASTM F 606/F606M. Testing is not required for any mechanical test attribute that does not have acceptance criteria for that attribute. Non-ferrous screws shall be excluded from the reduction in area requirement. For fasteners that are too short for preparing a machined test specimen in accordance with ASTM F606/F606M to support axial testing, longer length fasteners from the same lot, see 4.2.1, shall be produced to support turning down the body. The maximum length restriction per 4.2.1 for these longer fasteners does not apply.

4.3.1.3 Wedge tension test. When wedge tensile testing is required by Table VII, wedge testing shall be performed in accordance with ASTM F606/F606M, with wedge angles in accordance with Table X. Samples that have been yield tested shall not be used for wedge tensile testing. Samples that were used for proof testing may be used for wedge tensile testing, provided the fastener passed the proof test. Flat countersunk head and button head cap screws shall be exempt from the wedge tensile test.



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Table X Wedge Angles for Type VI Socket head cap screws

| Material | Size | Wedge Angle /1 | |
|---|---|---|---|
| | | Body Lengths greater than 2D | Body Lengths 2D or less or Threaded to the Head /2 |
| Alloy Steel | 0.112-0.50 incl. | 10 | 6 |
| | 0.625-0.750 incl. | 8 | 6 |
| | 0.875 and larger | 6 | 4 |
| CRES and Heat and Corrosion Resistant Steel | 0.112-0.750 incl. | 10 | 6 |
| | Over 0.750 | 6 | 4 |
| Nonferrous (except titanium) | 0.112-0.750 incl. | 10 | 6 |
| | Over 0.750 | 6 | 4 |
| Titanium | 0.112-0.750 incl. | 6 | 4 |
| | Over 0.750 | 4 | 4 |

Note: 1. Wedge angles listed are minimum values.
Note: 2. Socket head cap screws are defined as threaded to the head if the body length is two nominal diameters or less.

4.3.2 Metallographic test. Samples selected as specified in 4.2.5 shall be subjected to a metallographic test conducted in accordance with ASTM A574 or ASTM F835 to determine conformance with 3.6.

4.3.3 Protective coating or surface treatment tests. Protective coating or surface treatment tests shall be conducted in accordance with the applicable specification as specified in 3.3. Electroplated screws shall be subjected to the hydrogen embrittlement relief requirement and test specified in the applicable electrodeposited plating or coating specification specified in 3.3.

4.3.4 Chemical Analysis. Samples selected as specified in 4.2.7 shall be subjected to a chemical analysis in accordance with ASTM E1282. When a mill certificate covering chemical of utilized raw materials can be furnished, it will be acceptable in lieu of the chemical analysis of the finished part required by 4.2.7 (see 6.2)

4.3.5 Discontinuity inspection test. Samples selected as specified in 4.2.6 shall be subjected to liquid penetrant testing in accordance with ASTM E1417/E1417M, Type I or Type II, Method A, B or D or magnetic particle testing in accordance with ASTM E1444/E1444M inspection methods as applicable to meet the requirements of 3.7. The samples shall be metallurgically examined within a range from 15 to 200X magnification to determine the extent and depth of the discontinuities. NAVSEA Technical Publication T9074-AS-GIB-010/271 may be substituted for ASTM E1417/E1417M for liquid penetrant testing or E1444/E1444M for magnetic particle testing. QQ-N-286 Nickel-Copper-Aluminum alloy fasteners shall be inspected in accordance with T9074-AS-GIB-010/271.

4.3.6 Banded structure inspection. When a banded structure is observed of questionable acceptability, a minimum of 3 Knoop micro-hardness readings per test, in



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accordance with ASTM E384, shall be taken on the most predominate bands. The test load shall be 500 g, unless extremely thin bands are observed that will not permit an accurate 500 g test, in which case it is permissible to reduce the test load to as low as 50 g to obtain an accurate reading. Individual readings shall not exceed 470 HK. One test shall be conducted per heat.

4.3.7 Rounding of test results. When a rounding method is indicated, results of mechanical and chemistry tests shall be rounded in accordance with the rounding method of ASTM E29. Results of the chemistry analysis shall be rounded to the number of significant digits of the applicable test requirement. Tensile strength test values of up to 50,000 psi shall be rounded to the nearest 100 psi; test values over 50,000 psi and less than 100,000 psi shall be rounded to the nearest 500 psi; and test values over 100,000 psi shall be rounded to the nearest 1,000 psi. Elongation and reduction of area test values of up to 10 percent shall be rounded to the nearest 0.5 percent and test values over 10 percent shall be rounded to the nearest 1 percent.

5. Packaging.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. This specification covers externally threaded fasteners, which are designed to be driven into tapped holes with a hexagon or spline key or setter, in applications which require high strength precision products, compactness of design of the assembled parts, used in confined spaces or sinking of heads below surfaces into holes which fit the head.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Fastener type, material grade and condition (if applicable), size and length or standard part number (see 1.2, 1.2.1, 1.2.2 and 3.1).
c. Title, number and date including part number specified in accordance with military document (see 6.3).
d. Material certification, if required (see 3.1 and 4.3.4).
e. Protective coating or surface treatment, if required (see 3.3)
f. Cross drilling of cylindrical heads, if required (see 3.5.2).
g. Thread series and class (see 3.5.4.3).

6.3 Military Procurement. Cadmium plate fasteners are for replacement parts use only.



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NAS1351 – Screw, Cap, Socket Head, Undrilled and Drilled, Plain and Self-Locking, Alloy Steel and Corrosion-Resisting Steel and Heat-Resistant Steel, UNRF-3A.
NAS1352 – Screw, Cap, Socket Head, Undrilled and Drilled, Plain and Self-Locking, Alloy Steel and Corrosion-Resistant Steel and Heat-Resistant Steel, UNRC-3A and UNRC-2A.
NASM16995 – Screw, Cap, Socket Head-Hexagon, Corrosion Resistant Steel, UNC-3A
NASM16996 – Screw, Cap, Socket Head-Hexagon, Corrosion Resistant Steel, UNF-3A
NASM21262 – Screws, Self Locking, 250° F, Cylindrical Head Hexagonal Wrenching Socket, Alloy Steel, 160 KSI FTU (Socket Head Cap Screws)
NASM21295 – Screws, Self Locking, 250° F, Cylindrical Head, Hexagonal Wrenching Socket, CRES 80 KSI FTU (Socket Head Cap Screws)
NASM24667 – Screw, Cap, Socket Head, Flat Countersunk, 82 deg., Alloy Steel, UNC-3A
NASM24671 – Screw, Cap, Socket-Head-Flat Countersunk, 82 deg., Corrosion-Resisting Steel, UNC-3A
NASM24673 – Screw, Cap, Socket Head, Hexagon, Drilled Corrosion-Resisting Steel (UNF-3A)
NASM24674 – Screw, Cap, Socket Head, Hexagon, Drilled Corrosion Resistant Steel (UNC-3A)
NASM24677 – Screw, Cap, Socket Head, Hexagon, Drilled Alloy Steel, (UNC-3A)
NASM24678 – Screw, Cap, Socket Head, Hexagon, Drilled, Alloy Steel, (UNF-3A)
MS24672 – Screw, Cap, Socket Head, Flat Countersunk, Corrosion Resisting Steel, UNF-3A
MS35455 – Screw, Cap, Socket Head-Hexagon, Alloy Steel, Uncoated, UNC-3A
MS35456 – Screw, Cap, Socket Head-Hexagon, Alloy Steel, Uncoated, UNF-3A (Inactive for New Design)
MS35457 – Screw, Cap, Socket Head Hexagon, Alloy Steel, Cadmium or Zinc, UNC-3A (Inactive for New Design)
MS35458 – Screw, Cap, Socket Head-Hexagon, Alloy Steel, Cadmium or zinc, UNF-3A (Inactive for New Design)
MS35459 – Screw, Cap, Socket Head-Hexagon, Alloy Steel, Phosphate, UNC-3A (Inactive for New Design)
MS35460 – Screw, Cap, Socket Head-Hexagon, Alloy Steel, Phosphate, UNF-3A (Inactive for New Design)
MS35461 – Screw, Cap, Socket Head-Hexagon, Steel, Corrosion Resisting, Passivated, UNC-3A (Inactive for New Design)
MS51480 – Screw, Cap, Socket Head, Alloy Steel, UNF-3A (Inactive for New



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Design)
MS51484 – Screw, Cap, Socket Head, Alloy Steel, UNC-3A (Inactive for New Design)

6.3.1 Type I screws (1936 series). Type I screws (1936 series) are obsolete for new design. For procurement, it is recommended that the design be evaluated for use with type VI as replacement.

6.3.2 Types III, V, VII and IX screws. Types III, V, VII and IX are deleted from previous revisions of this specification. For procurement it is recommended that the design be evaluated for use with type IV, VI and VII hexagon as replacement.

6.3.3 Hazardous Materials. Cadmium coatings and coatings containing hexavalent chromium are obsolete and not to be used for new design.

6.4 Definitions of discontinuities.

6.4.1 Crack. A crack is a clean crystalline break passing through the grain or grain boundary without inclusion of foreign elements.

6.4.2 Seam. A seam is an elongated discontinuity in the metal caused by a defect which has been closed by rolling or forging mechanically, but not united.

6.4.3 Inclusion. An inclusion is particles of non-metallic impurities, usually oxides, sulfides, silicates and such, which are mechanically held in steel during solidification.

6.4.4 Lap. A lap is a surface defect appearing as a seam caused by folding over metal, fins or sharp corners, and then rolling or forging, but not welding them into the surface.

6.4.1 Nicks, Pits. Nicks and pits are a depression or indentation in the surface of the metal.

6.5 Subject term (key word) listing.

Button head cap screw
Cylindrical head cap screw
Flat countersunk cap screw
Socket head cap screw

6.6 Supersession data. This specification includes the requirements of FF-S-86E dated May 29, 1987 and Amendment 4 dated January 16. 1991, FF-S-86D dated 15 June 1971 and Amendment -1 dated 6 November 1972. Cross reference data between the types of screws covered by this specification and the types and styles of the preceding specifications, FF-S-86 dated 4 November 1953, FF-S-86A dated 16 May 1961, FF-S-86B dated 23 April 1964, FF-S-86C dated 17 October 1967, and FF-S-86D dated 15 June 1971, are as follows:



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Table XI Supersession Reference Table

| FF-S-86 | FF-S-86A | FF-S-86B, FF-S-86C, FF-S-86D, FF-S-86E | FF-S-86F, FF-S-86G |
|---|---|---|---|
| None | Type V | Type V | None 2/ |
| None | Type VI, 1960 Series | Type VI | Type VI |
| None | Type VII, 1960 Series | Type VII | None 2/ |
| None | None | Type VIII | Type VIII |
| None | None | Type IX | None 2/ |

1/ These specifications do not cover Type II, Style 11 of FF-S-86 since that type of fastener is an internal wrenching bolt.
2/ This specification does not cover Types I, III, V, VII and IX of FF-S-86E as these are either 1936 series or have an internal multiple spline socket head configuration. See 6.3.1 and 6.3.2.



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Table XII. Yield and tensile loads of corrosion resistant and heat and corrosion resistant steel socket head screws – coarse threads

| Nominal Diameter | Threads per Inch | Tensile Stress Area | Corrosion Resistant Steel | | Heat and Corrosion Resistant Steel | |
|---|---|---|---|---|---|---|
| | | | Yield Strength | Tensile Load | Yield Strength | Tensile Load |
| | | | Lbs. min. 3/ | Lbs. min. 3/ | Lbs. min. 3/ | Lbs. min. 3/ |
| 0.0730 | 64 | 0.00262 | 79.0 | 210 | 316 | 420 |
| 0.0860 | 56 | 0.00370 | 111.0 | 296 | 444 | 592 |
| 0.0990 | 48 | 0.00486 | 146.0 | 390 | 584 | 780 |
| 0.1120 | 40 | 0.00603 | 181.2 | 484 | 725 | 965 |
| 0.1380 | 32 | 0.00909 | 272.5 | 725 | 1,090 | 1,455 |
| 0.1640 | 32 | 0.01401 | 420 | 1,120 | 1,680 | 2,240 |
| 0.1900 | 24 | 0.01753 | 525 | 1,400 | 2,100 | 2,805 |
| 0.2500 | 20 | 0.03182 | 954 | 2,540 | 3,820 | 5,090 |
| 0.3125 | 18 | 0.05243 | 1,572 | 4,200 | 6,300 | 8,390 |
| 0.3750 | 16 | 0.07749 | 2,325 | 6,200 | 9,300 | 12,400 |
| 0.4375 | 14 | 0.10631 | 3,190 | 8,500 | 12,750 | 17,010 |
| 0.5000 | 13 | 0.14190 | 4,260 | 11,350 | 17,000 | 22,705 |
| 0.5625 | 12 | 0.18194 | 5,460 | 14,550 | 21,800 | 29,110 |
| 0.6250 | 11 | 0.22600 | 6,780 | 18,100 | 27,100 | 36,160 |
| 0.7500 | 10 | 0.33446 | 10,020 | 26,700 | 40,100 | 53,515 |
| 0.8750 | 9 | 0.46173 | 12,020 | 32,300 | 55,400 | 73,875 |
| 1.0000 | 8 | 0.60575 | 15,750 | 42,400 | 72,700 | 96,920 |
| 1.1250 | 7 | 0.76328 | 19,850 | 53,400 | 91,600 | 122,125 |
| 1.2500 | 7 | 0.96911 | 25,200 | 67,800 | 116,300 | 155,060 |
| 1.3750 | 6 | 1.15488 | 30,050 | 80,900 | 138,600 | 184,780 |
| 1.5000 | 6 | 1.40525 | 36,500 | 98,400 | 168,600 | 224,840 |

1/ Do not use for new design.
2/ $A_s = 3.1416[(D_2/2) - (3H/16)]^2$ or $A_s = 0.7854[D- 0.9743/(1/P)]^2$ as specified in ASME B1.1
3/ Values provided are for cylindrical head configuration. To allow for the critical nature of flat countersunk head and button head fastener configurations, the load shall be considered acceptable provided the full size fastener test meets 80 percent of the load (Types IV and VIII).
4/ Yield and tensile strength requirements were calculated by multiplying the yield and tensile strength minimum allowable stresses (of the corresponding machined specimen for the finished screw) by the tensile strength area of the threads.



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Table XIII. Yield and tensile loads of corrosion resistant and heat and corrosion resistant steel socket head cap screws – fine threads for full-sized specimens only

| Nominal size | Threads per inch | Tensile stress area 2/ | Corrosion Resistant Steel | | Heat and Corrosion Resistant Steel | |
|---|---|---|---|---|---|---|
| | | | Yield Strength | Tensile Load | Yield Strength | Tensile Load |
| | | | Lbs. min. 3/ | Lbs. min. 3/ | Lbs. min. 3/ | Lbs. min. 3/ |
| 0.0600 | 80 | 0.00180 | 54.0 | 144 | 216 | 288 |
| 0.0730 | 72 | 0.00278 | 83.4 | 222 | 334 | 444 |
| 0.0860 | 64 | 0.00393 | 118.2 | 316 | 472 | 630 |
| 0.0990 | 56 | 0.00523 | 157.0 | 418 | 628 | 836 |
| 0.1120 | 48 | 0.00660 | 198.5 | 530 | 794 | 1,060 |
| 0.1250 | 44 | 0.00831 | 249.5 | 665 | 998 | 1,330 |
| 0.1380 | 40 | 0.01014 | 304.5 | 810 | 1,220 | 1,620 |
| 0.1640 | 36 | 0.01473 | 442 | 1,180 | 1,770 | 2,360 |
| 0.1900 | 32 | 0.01999 | 600 | 1,600 | 2,400 | 3,200 |
| 0.2500 | 28 | 0.03637 | 1,092 | 2,910 | 4,350 | 5,820 |
| 0.3125 | 24 | 0.05807 | 1,740 | 4,640 | 6,950 | 9,300 |
| 0.3750 | 24 | 0.08783 | 2,635 | 7,020 | 10,500 | 14,050 |
| 0.4375 | 20 | 0.11872 | 3,560 | 9,500 | 14,200 | 19,000 |
| 0.5000 | 20 | 0.15995 | 4,800 | 12,800 | 19,200 | 25,600 |
| 0.5625 | 18 | 0.20298 | 6,090 | 16,250 | 24,400 | 32,500 |
| 0.6250 | 18 | 0.25596 | 7,680 | 20,500 | 30,700 | 41,000 |
| 0.7500 | 16 | 0.37296 | 9,700 | 26,100 | 44,800 | 59,700 |
| 0.8750 | 14 | 0.50947 | 13,240 | 35,600 | 61,100 | 81,400 |
| 1.0000 | 12 | 0.66304 | 17,250 | 46,400 | 79,600 | 106,100 |
| 1.1250 | 12 | 0.85572 | 22,250 | 59,900 | 102,700 | 137,000 |
| 1.2500 | 12 | 1.07295 | 27,900 | 75,100 | 128,800 | 171,700 |
| 1.3750 | 12 | 1.31471 | 34,200 | 92,100 | 157,800 | 210,400 |
| 1.5000 | 12 | 1.58102 | 41,100 | 110,700 | 189,700 | 253,000 |
| 1.0000 | 14NS | 0.67989 | 17,700 | 47,500 | 81,500 | 108,600 |

1/ Do not use for new design.
2/ $A_s = 3.1416[(D_2/2) - (3H/16)]^2$ or $A_s = 0.7854[D - 0.9743/(1/P)]^2$ as specified in ASME B1.1
3/ To allow for the critical nature of flat countersunk head and button head fastener configurations, the load shall be considered acceptable provided the fastener meets 80 percent of the load (Types IV and VIII).
4/ Yield and tensile strength requirements were calculated by multiplying the yield and tensile strength minimum allowable stresses (of the corresponding machined specimen for the finished screw) by the tensile strength area of the threads.



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Table XIV. Mechanical Properties of non-ferrous screws for full-sized specimens only

| Material | Applicable Document | Composition, class or alloy number | Condition or temper of bar | Ultimate tensile strength PSI min, unless shown as a range | Yield Strength PSI min 2/ | Elongation percent, min 1/ | Hardness 4/ |
|---|---|---|---|---|---|---|---|
| Manganese Bronze | ASTM F468 or ASTM B138/B138M | Cu 675 UNS C67500 | O60 - soft anneal | 55,000 - 85,000 | 22,000 2/ | 20 | HRB60-90 |
| Nickel Aluminum Bronze | ASTM B150/B150M | UNS C63200 | TQ50 - quench hardened and temper | 90,000 | 50,000 2/ | 15 | HRB96 min |
| Phosphor Bronze | ASTM F468 or ASTM B139/B139M | Cu510 UNS C51000 | H04/Hard | 80,000 (dia. Under .250) | 35,000 2/ | Not Specified | HRB60-95 |
| | | | | 70,000 (Dia. .250 to .500) | | 13 | |
| | | | | 60,000 (Dia. = over .500 to 1.000) | | 15 | |
| | | | | 55,000 (Dia. 1.000 to 3.000) | | 18 | |
| | | | | 50,000 (Dia. = over 3.000) | | 18 | |
| Silicone Bronze | ASTM F468 or ASTM B98/B98M or | Cu651 UNS C651000 | 04H - Hard | 60,000 | 40,000 2/ | 10 | HRB60-95 |
| | ASTM B99/B99M | UNS C65100 | 01H - quarter hard | | | | |
| Naval Brass | ASTM F468 or ASTM B21/B21M | Cu464 UNS C46400 | H02 - Half-hard or O50 Light Annealed | 50,000 | 27,000 2/ | 22 (Dia. = .500 & under 25 (Dia. = over .500) | HRB55-75 |
| Nickel Copper Alloy | ASTM F468 or QQ-N-281 | Ni400 Ni405 Class A UNS N04400 | Hot Finished or Cold Drawn | 80,000 | 40,000 2/ | 20 | HRB75 - HRC25 |
| Nickel Copper Aluminum Alloy | QQ-N-286 | Ni500 UNS N05500 | Annealed and Age Hardened | 130,000 | 90,000 (Dia. = under 1.000) 85,00 (Dia. = 1.000 and over) 2/ | 20 | HRC24-35 |
| Inconel 625 5/ | ASTM F468 or ASTM B446, Grade 1 | Ni625 UNS N06625 | | 120,000 | 60,000 | 30 | HRB85-HRC35 |
| Inconel 686 | ASTM F468 or ASTM B574 | Ni686 UNS N06686 | Hor or Cold Worked and Annealed | 120,000 - 165,000 | 110,000 | 12 | HRC25-36 |
| Titanium | ASTM F468 or ASTM B348 | Ti23 UNS R56407 | Hot Worked or Cold Finished | 120,000 - 165,000 | 110,000 | 12 | HRC25-36 |

1/ In 4D gage length.
2/ 0.2 percent offset unless otherwise cited in referenced standard.



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3/ For arbitration purposes, Ultimate Tensile Strength takes precedence over Hardness.
4/ Due to differences in mechanical properties for non-ferrous materials between Industry Standards and FF-S-86, new designs are encouraged to use SAE J2295.
5/ Finished fasteners shall be furnished in the annealed condition.

Table XV. Mechanical Properties for Type VI Socket Head Cap Screws (Finished Products)

| Material Grade | Condition | Nominal Diameter (in.) | Ultimate Tensile Strength (KSI) Minimum | Yield Tensile Strength Minimum | Proof Strength Minimum | Hardness Minimum (Note 1) |
|---|---|---|---|---|---|---|
| AS | Not Specified | #10 - 0.500 | 180 | | 140 | 39 - 45 HRC (Note 1) |
| | Not Specified | Over 0.500 | 170 | | 135 | 37 - 45 HRC (Note 1) |
| SS300 | Annealed (A) | #10 - 0.625 | 80 | 30 | | 80 |
| | | Over 0.625 | 70 | 26 | | 74 |
| | Cold Worked (CW) | #10 - 0.625 | 102 | 65 | | 96 |
| SS316 | Annealed (A) | #10 - 0.625 | 80 | 30 | | 80 |
| | | Over 0.625 | 70 | 26 | | 74 |
| | Cold Worked (CW) | #10 - ).625 | 102 | 65 | | 96 |
| A286 | Not Specified | All | 160 | 120 | | 33- 42HRC |
| 675A | O60 - Soft Anneal | All | 55 -85 | 25 | | 60 - 90HRB |
| 632A | TQ50 - Quench Hardened and Tempered | All | 90 | 50 | | 96HRB |
| 510A | H04 - Hard | #10 - 0.250 | 80 | 35 | | 60 - 95HRB |
| | | Over 0.250 - 0.500 | 70 | 35 | | |
| | | Over 0.500 - 1.00 | 60 | 35 | | |
| | | Over 1.00 - 3.00 | 55 | 35 | | |
| | | Over 3.00 | 50 | 35 | | |
| 651A | 04H - Hard or 01H - Quarter Hard | All | 60 | 40 | | 60-95HRB |
| 464A | H02 - Half Hard or O50 - Light Annealed | All | 50 | 27 | | 55-75HRB |
| N400 | Hot Finished or Cold Drawn | All | 80 | 40 | | 75HRB - 25HRC |
| N500 | Annealed and Age Hardened | Under 1.00 | 130 | 90 | | 24 - 35HRC |
| | | 1.00 and Over | 130 | 85 | | |
| 625A | Annealed | All | 120 | 60 | | 24 - 35HRC |
| 686A | Hot or Cold Worked and Annealed | All | 120 (165 Max) | 85 | | 21 -45HRC |
| T23 | Hot Worked or Cold Finished | All | 120 (165 Max) | 110 | | 25 - 36HRC |

Note 1: Screws in accordance with 3.3.2 or 3.3.3 shall have a maximum hardness of 43HRC. Minimum hardness shall be as specified in Table IX – XI.



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| Table XVI. Mechanical Properties for Type IV Flat Countersunk and Type VIII Socket Button Head Cap Screws (Finished Products) | | | | | | |
|---|---|---|---|---|---|---|
| Material Grade | Condition | Nominal Diameter (in) | Ultimate Tensile Strength (KSI) Minimum | Yield Tensile Strength Minimum | Proof Strength Minimum (See Note 1) | Hardness Minimum (See Note 2) |
| AS | Not Specified | #10 - 0.500 | 144 | | 122 | 39 - 45HRC (Note 3) |
| | Not Specified | Over 0.500 | 136 | | 108 | 37 - 45HRC (Note 3) |
| SS300 | Annealed (A) | #10 - 0.625 | 64 | 24 | 21.6 | 80 |
| | | Over 0.625 | 56 | 20.8 | 18.7 | 74 |
| | Cold Worked (CW) | #10 - 0.625 | 82 | 52 | 46.8 | 96 |
| SS316 | Annealed (A) | #10 - 0.625 | 64 | 24 | 21.6 | 80 |
| | | Over 0.625 | 56 | 20.8 | 18.7 | 74 |
| | Cold Worked (CW) | #10 - 0.625 | 82 | 52 | 46.8 | 96 |
| A286 | Not Specified | All | 128 | 96 | 86.4 | 33 - 42HRC |
| 675A | O60 - Soft Anneal | All | 44-68 | 20 | 18 | 60-90HRB |
| 632A | TQ50 - Quench Hardened and Tempered | All | 72 | 40 | 36 | 96HRB |
| 510A | H04 - Hard | #10 - 0.250 | 64 | 28 | 25.2 | 60-95HRB |
| | | Over 0.250 - 0.500 | 56 | 28 | 25.2 | |
| | | Over 0.500 -1.00 | 48 | 28 | 25.2 | |
| | | Over 1.00 - 3.00 | 44 | 28 | 25.2 | |
| | | Over 3.00 | 40 | 28 | 25.2 | |
| 651A | 04H - Hard or 01H - Quarter Hard | All | 48 | 32 | 28.8 | 60-95HRB |
| 464A | H02 - Half Hard or O50 - Light Annealed | All | 40 | 21.5 | 19.5 | 55-75HRB |
| N400 | Hot Finished or Cold Drawn | All | 64 | 32 | 28.8 | 75HRB - 25HRC |
| N500 | Annealed and Age Hardened | Under 1.00 | 104 | 72 | 64.8 | 24-35HRC |
| | | 1.00 and over | 104 | 68 | 61.2 | |
| 625A | Annealed | All | 96 | 48 | 43.2 | 24-35HRC |
| 686A | Hot or Cold Worked and Annealed | All | 96 (132 Max) | 68 | 61.2 | 21-45HRC |
| T23 | Hot Worked or Cold Finished | All | 96 (132 Max) | 88 | 79.225-36HRC | |

Note 1: Yield strength is listed for informational purposes only. The critical nature of these fastener head designs may allow for yielding to occur outside of the grip defined in ASTM F606/F606M during tensile testing. Therefore, tensile test data shall not be used to certify the yield strength of Type IV and VIII fasteners unless a test method is agreed upon by the user and manufacturer.
Note 2: Screws electroplated in accordance with 3.3.2 or 3.3.3 shall have a maximum hardness of 43HRC. Minimum hardness to be as specified in Tables XV – XVII.



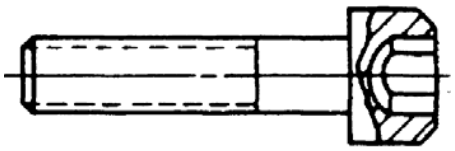
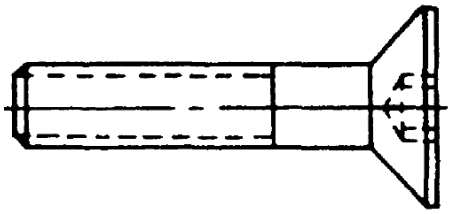
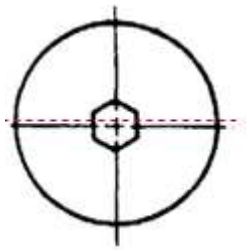
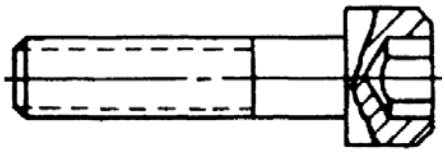
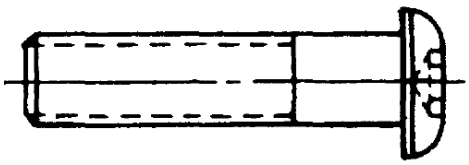
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| Table XVII. Mechanical Properties for Machined Specimens | | | | | | |
|---|---|---|---|---|---|---|
| Material Grade | Condition | Nominal Diameter (in) | Ultimate Tensile Strength (KSI) Minimum | Yield Tensile Strength Minimum | Hardness Minimum (Note 1) | Elongation in 4 Diameters Minimum |
| AS | Not Specified | #10 - 0.500 | 180 | 155 | 39-45HRC (Note 2) | 12 Percent |
| | Not Specified | Over 0.500 | 170 | 150 | 37-45HRC (Note 2) | 12 Percent |
| SS300 | Annealed | #10 - 0.625 | 80 | 30 | 80HRB | 10 Percent |
| | | Over 0.625 | 70 | 26 | 74HRB | 20 Percent |
| | Cold Worked (CW) | #10 - 0.625 | 102 | 65 | 96HRB | 25 Percent |
| SS316 | Annealed | #10 - 0.625 | 80 | 39 | 80HRB | 10 Percent |
| | | Over 0.625 | 79 | 26 | 74HRB | 20 Percent |
| | Cold Worked (CW) | #10 - 0.625 | 102 | 65 | 96HRB | 25 Percent |
| A286 | Not Specified | All | 160 | 120 | 33-42HRC | 12 Percent |
| 675 | O60 - Soft Anneal | All | 55 | 25 | 60-90HRB | 20 Percent |
| 632A | TQ50 - Quench Hardened and Tempered | All | 90 | 50 | 96HRB | 15 Percent |
| 510A | H04 - Hard | #10 - 0.250 | 80 | 35 | 60-95HRB | Not Specified |
| | | Over 0.250 - 0.500 | 70 | 35 | | 13 Percent |
| | | Over 0.500 - 1.00 | 60 | 35 | | 15 Percent |
| | | Over 1.00-3.00 | 55 | 35 | | 18 Percent |
| | | Over 3.00 | 50 | 35 | | 18 Percent |
| 651A | 04H - Hard or 01H - Quarter Hard | All | 60 | 40 | 60-95HRB | 10 Percent |
| 464A | H02 - Half Hard or O50 - Light Annealed | All | 50 | 15 | 55-75HRB | #10 - 0.500 = 10 Percent |
| | | | | | | Over 0.500 = 25 Percent |
| N400 | Hot Finished or Cold Drawn | All | 80 | 40 | 75HRB - 25HRC | 20 Percent |
| N500 | Annealed and Age Hardened | Under 1.00 | 130 | 90 | 24-35HRC | 20 Percent |
| | | 1.00 and Over | 130 | 85 | | |
| 625A | Annealed | All | 120 | 60 | 24-35HRC | 30 Percent |
| 686A | Hot or Cold Worked and Annealed | All | 120 (165 Max) | 85 | 21-45HRC | 20 Percent |
| T23 | Hot Worked or Cold Finished | #10 - 2.00 | 120 (165 Max) | 110 | 25-36HRC | 12 Percent |
| | | Over 2.00 | 115 (165 Max) | 105 | 25-36HRC | 12 Percent |

Note 1: Screws electroplated in accordance with 3.3.2 or 3.3.3 shall have a maximum hardness of 43HRC. Minimum hardness to be as specified in Tables XV – XVII.



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| Types | Drives | Heads |
|---|---|---|
| Type I |  CYLINDRICAL |  |
| Type IV Hexagon |  FLAT |  |
| Type VI, Hexagon |  CYLINDRICAL |  |
| Type VIII Hexagon ~~1960 Series~~ 1/ |  BUTTON |  |

1/ Obsolete for new design (see 6.3.1). For procurement, it is recommended that the design be evaluated for use with type VI as replacement. Internal multiple spline fasteners should be replaced with hex configuration for types III, V, VIII and IX of the same head configuration.

Figure 7. Socket-head screw types.



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Military Interest:

Custodians:
Army – AR
Navy – SH

Preparing Activity:
DLA – IS

(Project: 5305-2020-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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